Table 13 – continued from previous page

Model	Method	LongBench MAE	ZeroSCROLLS MAE
	SSJF-MC	930.60	613.32
	S3	254.55	119.09
	PiA	153.00	161.45
	TPV	790.71	361.45
	TRAIL	**143.47**	152.36
	LTR-C	153.51	94.95
	Ours	**132.43**	**78.83**
Qwen2.5 -7B -Instruct	SSJF-Reg	201.49	100.93
	SSJF-MC	832.60	182.08
	S3	223.64	**100.00**
	PiA	256.32	154.24
	TPV	676.28	391.57
	TRAIL	**150.46**	117.91
	LTR-C	158.45	100.29
	Ours	**114.97**	**93.43**
Llama3.2 -1B -Instruct	SSJF-Reg	290.54	**57.40**
	SSJF-MC	431.19	72.24
	S3	450.91	73.64
	PiA	**145.61**	142.41
	TPV	686.38	337.95
	TRAIL	161.89	128.82
	LTR-C	307.63	**50.59**
	Ours	**93.36**	59.19
Llama3.2 -3B -Instruct	SSJF-Reg	311.58	**75.09**
	SSJF-MC	387.15	82.50
	S3	441.82	**77.27**
	PiA	**142.32**	145.61
	TPV	1079.16	388.35
	TRAIL	172.45	114.79
	LTR-C	504.68	131.10
	Ours	**102.47**	79.03

Table 14: Performance of length prediction methods on Reasoning Scenario.

Model	Method	IFeval MAE
Qwen2.5 -0.5B -Instruct	SSJF-Reg	**149.27**
	SSJF-MC	483.46
	S3	200.92
	PiA	260.06

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Table 14 – continued from previous page

Model	Method	MAE
	TPV	597.00
	TRAIL	146.24
	LTR-C	178.28
	Ours	**145.81**
Qwen2.5 -1.5B -Instruct	SSJF-Reg	**133.99**
	SSJF-MC	597.21
	S3	180.73
	PiA	265.15
	TPV	584.08
	TRAIL	**138.01**
	LTR-C	139.02
	Ours	146.48
Qwen2.5 -3B -Instruct	SSJF-Reg	**129.00**
	SSJF-MC	812.80
	S3	223.85
	PiA	254.57
	TPV	621.72
	TRAIL	132.19
	LTR-C	145.53
	Ours	**111.23**
Qwen2.5 -7B -Instruct	SSJF-Reg	**120.52**
	SSJF-MC	599.25
	S3	168.81
	PiA	254.54
	TPV	466.56
	TRAIL	124.19
	LTR-C	134.55
	Ours	**119.60**

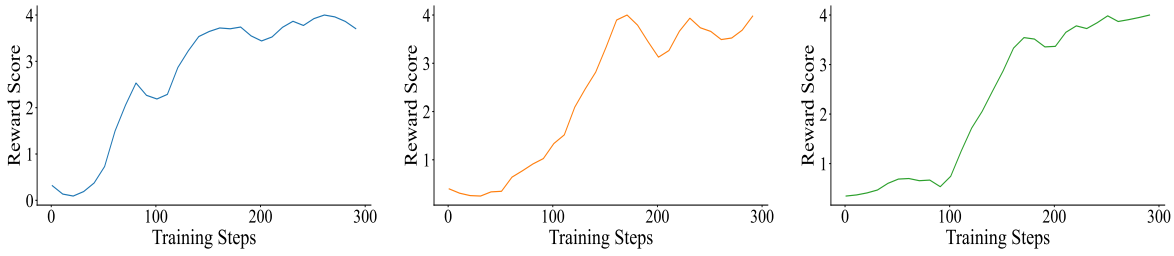
Table 15: Performance of length prediction methods on Reasoning Scenario.

Model	Method	MAE
DeepSeek-R1-Distill -Qwen-1.5B	SSJF-Reg	86.16
	SSJF-MC	90.36
	S3	164.22
	PiA	264.66
	TPV	373.03
	TRAIL	77.16
	LTR-C	**74.01**
	Ours	**71.59**
DeepSeek-R1-Distill -Qwen-7B	SSJF-Reg	57.67
	SSJF-MC	143.78
	S3	52.29
	PiA	254.57
	TPV	273.04
	TRAIL	58.28
	LTR-C	**52.28**
	Ours	**48.36**
DeepSeek-R1-Distill -Llama-8B	SSJF-Reg	72.00
	SSJF-MC	78.21
	S3	69.45
	PiA	272.16
	TPV	243.54
	TRAIL	**46.85**
	LTR-C	66.06
	Ours	**45.87**

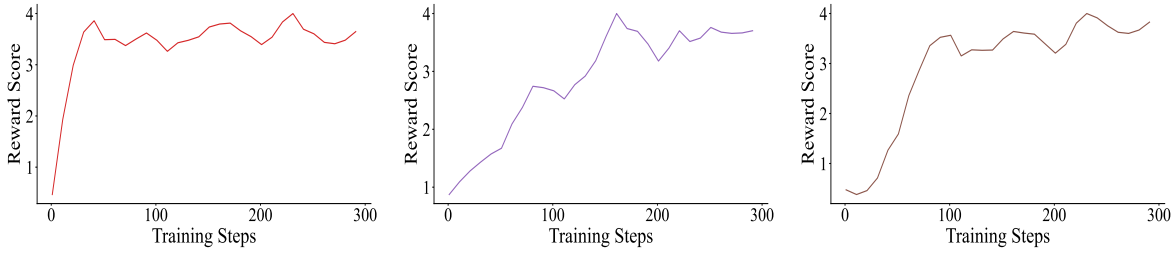
E.7 Visualization of the GRPO Training Process

To demonstrate the effectiveness of our GRPO training process, we present the reward curves from our experiments in Figure 8 through Figure 13. These visualizations cover the training process on six datasets: code execution prediction with CRUXEval (Figure 8), code generation with MBPP (Figure 9) and LiveCodeBench (Figure 12), and mathematical and scientific reasoning with MMLU-STEM (Figure 10), MATH (Figure 11), and GSM8K (Figure 13).

For each dataset, we trained multiple models, including four different sizes of Qwen2.5 and two sizes of Llama3.2. It should be noted that for smaller models, we use a larger batch size, which can result in a different number of training steps. As shown across all figures, the training process exhibits a stable and consistent improvement in rewards. This demonstrates that our GRPO implementation successfully optimizes model performance, regardless of the underlying architecture or the specific challenges of the task.

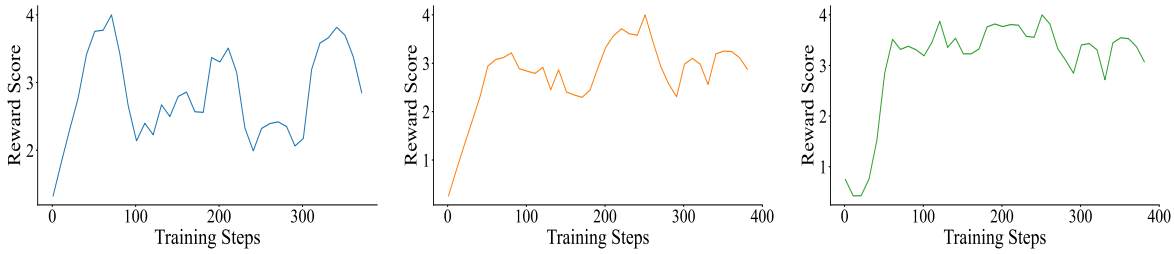


(a) Qwen2.5-0.5B (b) Qwen2.5-1.5B (c) Qwen2.5-3B

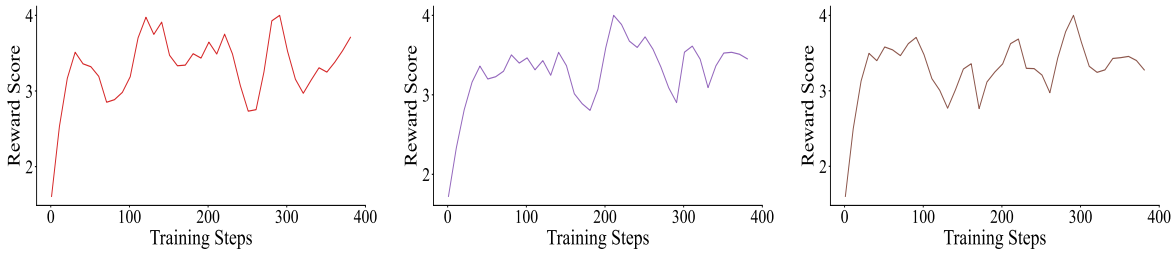


(d) Qwen2.5-7B (e) Llama3.2-1B (f) Llama3.2-3B

Figure 8: GRPO training process on the CRUXEval dataset.



(a) Qwen2.5-0.5B (b) Qwen2.5-1.5B (c) Qwen2.5-3B



(d) Qwen2.5-7B (e) Llama3.2-1B (f) Llama3.2-3B

Figure 9: GRPO training process on the MBPP dataset.